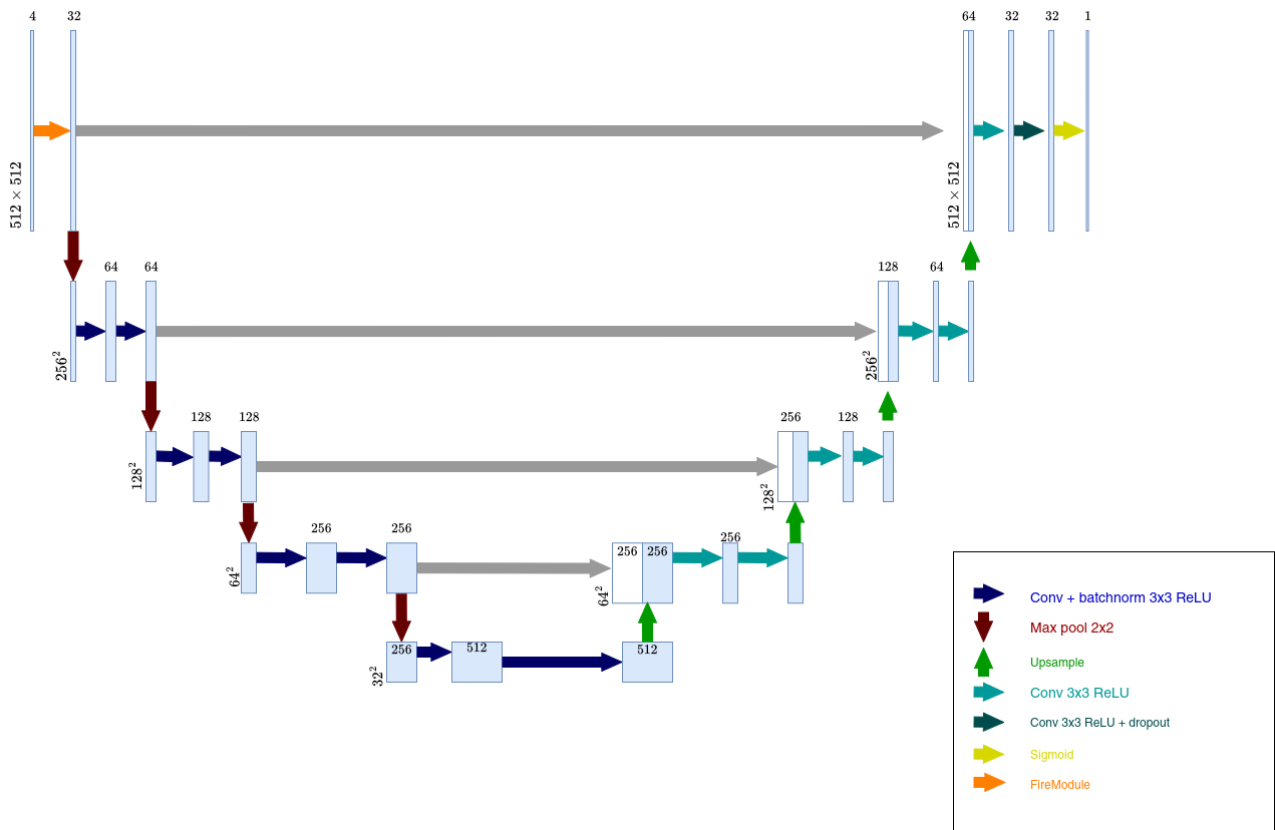


# Satellite Project: Cloud Masking

## Team Members

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## Project Pipeline



# PreProcessing

The preprocessing pipeline includes:

- Loading and organizing satellite images and their corresponding masks from TIFF files
- Converting GDAL datasets to NumPy arrays using `convert_to_nparr` function
- Splitting the dataset into training (60%), validation (20%), and test (20%) sets
- Applying data augmentation with random horizontal flips, vertical flips, and 45-degree rotations
- Using PyTorch's DataLoader for efficient batch processing

## Exploratory Data Analysis

- Dataset consists of 10,573 satellite images with corresponding cloud masks
- Images are in TIFF format with 4 spectral bands (shape: 4x512x512)
- Masks are binary (cloud vs. no-cloud) with shape 1x512x512
- Data distribution:
  1. Training: 6,343 images
  2. Validation: 2,115 images
  3. Test: 2,115 images
- Dataset was shuffled with a fixed random seed (42) for reproducibility

# Model Architecture

We implemented a modified U-Net architecture with SqueezeNet-inspired Fire modules:

- Encoder-decoder structure with skip connections
- Initial layer replaced with FireModule (squeeze-expand architecture)
- 4 levels of downsampling and upsampling
- Batch normalization and dropout ( $p=0.5$ ) for regularization
- Final sigmoid activation for binary classification
- Input channels: 4 (for the 4 spectral bands)
- Output channels: 1 (binary mask)

Key components:

- FireModule: Squeeze (1x1 conv) followed by parallel 1x1 and 3x3 expansions
- ConvBlock: Two convolutional layers with optional batch norm and dropout
- Transposed convolutions for upsampling

# Training Configuration

- Batch size: 16
- Learning rate: 0.001
- Number of epochs: 20
- Loss function: Binary Cross Entropy Loss (BCELoss)
- Optimizer: SGD
- Dropout probability: 0.5
- Training hardware: NVIDIA T4 GPU (via Kaggle)
- Early stopping based on validation Dice coefficient

# Training Results

The model showed consistent improvement during training:

- Initial training loss: 0.6924
- Final training loss: 0.5858
- Best validation Dice coefficient: 0.8562 (epoch 19)
- Training time: Approximately 4 hours for 20 epochs

# Evaluation Metrics

- Test Dice Coefficient: 0.8473
- Model performance was consistent between validation and test sets
- The Dice coefficient improved from 0.8027 in epoch 1 to 0.8562 in epoch 19
- Number of operations: 92.6 billion
- Model size: 7.7 million parameters

# Inference Pipeline

- Model loading and evaluation on test set
- Thresholding predictions at 0.5 for binary masks
- Batch processing of test images
- Generation of submission CSV file

# Future Work

Potential improvements:

- Experiment with different loss functions (Dice loss, Focal loss)
- Try more advanced architectures (Attention U-Net, ResUNet)